
PHASE 3 ACTIVITY SUMMARY

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1 Activity Overview

Our Phase 3 work addressed critical limitations in our Phase 2 submission: opaque decision-making relying on Whisper embeddings that conflated linguistic and paralinguistic signals, and poor generalizability due to shortcut learning. Most egregiously, our model was extremely confident that all Mandarin speakers had MCI, exploiting the Ye corpus sampling artifact rather than identifying genuine biomarkers.

Guiding Philosophy: Our approach applied “The Bitter Lesson” (Sutton) through **related data**, recognizing that while ADRD data is scarce (~2000 samples), “normal human language” and “healthy speech patterns” are solved problems. We leveraged Mozilla Common Voice for paralinguistic normalization and ChatGPT-4 for linguistic feature extraction.

Key Changes: Replaced XGBoost ensemble + Whisper with Hierarchical Bayesian Logistic Regression, eliminated Whisper dependency, separated prediction into three interpretable components (demographic, linguistic, paralinguistic), and added explicit corpus-level modeling to control rather than exploit sampling biases.

2 Data

We evaluated four external data sources with mixed results. Our most successful innovation was curated synthetic linguistic data using ChatGPT-4: 100 “Cookie Theft”-style scenarios paired with 66 linguistic ADRD markers generating healthy vs. symptomatic examples (cost <\$25, released on HuggingFace). This enabled quantification of previously difficult-to-measure symptoms without exposing patient data.

The Mozilla Common Voice dataset (filtered to English/Spanish speakers 50+, yielding 250 hours) provided our second breakthrough, allowing representation learning on GeMAPS features to model healthy speech baselines. Two approaches failed: text-to-speech normalization (unsolvable alignment problems) and YouTube ADRD content (abandoned due to privacy/licensing concerns).

3 Generalizability

3.1 Addressing Corpus Overfitting

Our hierarchical structure with corpus-specific varying intercepts/slopes and global priors enables explicit base rate adjustment for new populations. Linguistic effects apply identically across corpuses (universal biomarkers) while paralinguistic effects allow corpus-specific scaling. Unlike our Phase 2 black box, the Bayesian framework enables counterfactual queries: “How would this prediction change under different demographic assumptions?”

4 Methodology

4.1 Mathematical Framework

The model predicts binary outcome y_i following a Bernoulli distribution:

$$y_i \sim \text{Bernoulli}(p_i), \quad p_i = \text{logit}^{-1}(\mu_i) = \frac{1}{1 + e^{-\mu_i}}$$

The linear predictor combines effects from all features, where $j[i]$ is the corpus for individual i :

$$\mu_i = \underbrace{\alpha_{j[i]}}_{\text{VaryingIntercept}} + \underbrace{\sum_{k=1}^K \beta_{j[i],k} \cdot x_{i,k}^{(dem)}}_{\text{Demographic}} + \underbrace{\sum_{l=1}^L \gamma_{j[i],l} \cdot x_{i,l}^{(paralinguistic)}}_{\text{Paralinguistic}} + \underbrace{f(x_i^{(linguistic)})}_{\text{BART}}$$

Hierarchical Priors: Varying intercepts: $\alpha_j \sim \text{Normal}(0, \sigma_\alpha^2)$. Corpus-specific coefficients drawn from global distributions: $\beta_{j,k} \sim \text{Normal}(\mu_{\beta_k}, \sigma_{\beta_k}^2)$ with global priors $\mu_{\beta_k} \sim \text{Normal}(0, 1)$. BART function: $f(x^{(linguistic)}) = \sum_{t=1}^{150} g_t(x^{(linguistic)})$ using threshold-based decision trees.

4.2 Whisper Replacement Strategy

The counter-intuitive finding that Whisper-small outperformed Whisper-large suggested transcription accuracy was negatively correlated with paralinguistic feature extraction, motivating our complete departure from Whisper.

New Paralinguistic Pipeline: GeMAPS extraction (88 acoustic features) $\rightarrow$ representation learning on Common Voice time series $\rightarrow$ dual processing (XGBoost + fine-tuned embeddings) $\rightarrow$ corpus scaling.

4.3 Paralinguistic Encoder Architecture

We trained a hybrid CNN-Transformer encoder on Mozilla Common Voice GeMAPS time series using contrastive learning. The architecture consists of residual CNN blocks followed by a Transformer encoder with learnable [CLS] token.

Residual Blocks: Each block applies two 1D convolutions with batch normalization and ReLU activation:

$$h_l = \text{ReLU}(\text{BN}(W_2 \cdot \text{ReLU}(\text{BN}(W_1 \cdot h_{l-1}))) + h_{l-1})$$

where h_{l-1} is the input and skip connections handle dimension changes via 1x1 convolutions.

CNN Feature Extractor: Five residual blocks progressively increase channels: $88 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 512$, with 30% dropout between blocks.

Transformer Encoder: The CNN output is reshaped to $[batch, seq_len, 512]$, prepended with a learnable [CLS] token, and processed through 8 Transformer encoder layers with 16 attention heads and 2048 feedforward dimension.

Contrastive Learning: Two augmented views of each sequence are encoded to normalized embeddings $z_1, z_2 \in \mathbb{R}^{512}$. The contrastive loss maximizes similarity between positive pairs:

$$\mathcal{L} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{I}_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)}$$

where $\tau = 0.1$ is temperature and $\text{sim}(u, v) = u^T v / (\|u\| \|v\|)$.

The model was trained for 1500 epochs over 24 hours using aggressive augmentation (Gaussian noise, time masking, feature masking) to learn robust representations of healthy speech patterns.

New Linguistic Pipeline: Universal sentence encoder $\rightarrow$ 3-layer MLPs on synthetic data $\rightarrow$ feature selection (convergence + importance > 0.5) $\rightarrow$ BART integration.

4.4 Linguistic Classifier Training

For each of the 66 linguistic symptoms, we trained 3-layer MLP classifiers (512→256→2 neurons) on sentence embeddings from our synthetic dataset. Models were trained for 100 epochs with early stopping, and only symptoms with converged models showing importance >0.5 were retained for the final BART component.

Hardware: All model training was conducted locally using an RTX 4090 GPU with 128GB system RAM.

Software: The model fitting process relies on the Python ecosystem, primarily using PyMC3 and PyTorch.

BART Clinical Relevance: BART’s threshold-based trees mirror clinical symptom presentation, identifying cutoff points where markers become significant (e.g., “semantic approximations become concerning when circumlocutory descriptions exceed threshold frequency”) rather than assuming continuous relationships.

4.5 Expert Interpretability

Unlike Phase 2’s black box, our Bayesian framework supports: counterfactual queries, component decomposition (“Is prediction driven by linguistic, paralinguistic, or demographic factors?”), expert intervention (manual weight adjustment), and uncertainty quantification (confidence intervals vs. point predictions).

4.6 Methodological Influences

Our approach leveraged established ADRD research emphasizing separation of demographic, linguistic, and paralinguistic signals. Key literature insights: linguistic markers from ADRD research informed our 66-symptom synthetic generation; GeMAPS as fundamental paralinguistic tool providing standardized acoustic features; prior work on corpus effects guided hierarchical modeling.

Novel Contributions:

1. First LLM-generated synthetic data for ADRD linguistic feature extraction
2. Representation learning designed to separate linguistic from paralinguistic signals
3. Hierarchical Bayesian framework for corpus bias mitigation
4. Modular architecture: Clinicians can selectively adopt components based on clinical context
5. Publicly released GeMAPS encoder: Available for community use, extensible to 57 languages via Common Voice

The Phase 3 model demonstrates that principled signal separation, combined with appropriate external data integration, achieves improved generalizability and interpretability without sacrificing performance.

5 References

Sutton, R. S. (2019). The bitter lesson. Incomplete Ideas (blog). <http://www.incompleteideas.net/IncIdeas/BitterLesson.html>